Summary

Circadian rhythms are fundamental to organismal physiology and fitness. In plants and mammals, a transcription-translation feedback loop (TTFL) model has been proposed to generate the ~24-hr cell-autonomous circadian rhythm. It has been broadly accepted that the rhythmicity generated by the TTFL is propagated through transcript levels, protein levels, and protein activity, ultimately leading to rhythmicity in overall physiology. However, this model is challenged by recent proteomics findings, including low protein oscillation amplitudes and poor overlap between rhythmic transcripts and proteins. These findings prompted the development of the post-translational rhythmicity model, which suggests that the major TTFL output directly regulates protein activity post-translationally (e.g. through phosphorylation, sequestration). In this model, transcript-level oscillations observed in experiments are not the driver for the circadian rhythm. This raises the hypothesis that transcript oscillation might be an epiphenomenon of messenger RNA (mRNA) level homeostasis conferred by the TTFL output, to balance systemic rhythmic inputs. If this hypothesis is true, breaking the TTFL should lead to system-wide differential expression of genes that were rhythmic in the wild type. To test this hypothesis in mammals, time-course RNA-seq data from mouse mutants with a broken TTFL were selected, and a transcriptomics analysis pipeline was built to integrate rhythmicity detection and differential expression analysis. With a defined amplitude and fold-change threshold, no system-wide DE was observed across datasets. This result does not support the mRNA homeostasis hypothesis but revealed that high-amplitude rhythmic genes are more likely to exhibit differential expression.

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Abbreviations

Introduction

To maximise the chance of survival in Earth's 24-hour light-dark cycle, most organisms, from prokaryotes to plants and animals, have evolved an internal circadian clock as well as mechanisms to detect and integrate environmental light information (Paranjpe & Sharma, 2005). In mammals, the suprachiasmatic nucleus (SCN) in the brain was found to be the master circadian **pacemaker and** coordinator (Hastings et al., 2018) while the intrinsic circadian clock in **each** cell forms the fundamental basis of circadian rhythmicity (Welsh et al., 2004; Yoo et al., 2004).

Over the past 50 years, a transcriptional-translation feedback loop (TTFL) model of the mammalian circadian clock has been proposed and widely accepted (Takahashi, 2017), which involves a central TTFL (Fig. X). In this network, transcription factors Circadian Locomotor Output Cycles Kaput (CLOCK) and Brain and Muscle ARNT-like 1 (BMAL1 or ARNTL1) dimerise to activate the transcription of Period (PER, with paralogs Per1/2/3) and Cryptochrome (CRY, with paralogs CRY1/2). After being translated in the cytoplasm, PER and CRY form a complex with CK1, which phosphorylates PER. This complex then enters the nucleus, where it inhibits CLOCK:BMAL1 and hence its own transcription, forming a negative feedback loop. The timing of transcription, translation, post-translational modification and translocation altogether generates the core 24-hour rhythm. Various auxiliary loops have also been identified, involving such as nuclear receptors REV-ERBs and RORs and D-box binding protein (DBP) (Takahashi, 2017), to maintain the stability and robustness of the core rhythm. A functionally similar TTFL model has been proposed for the plant circadian clock - a network of feedback loops involving complexes such as **CCA1/LHY, PRR9/PRR7, PRR5/TOC1, and ELF4/LUX.**

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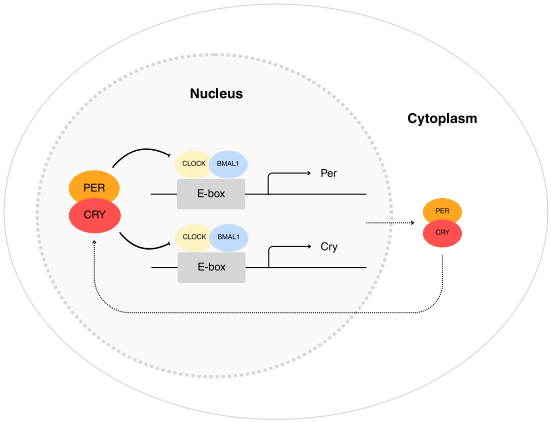

Fig. X: Simplified illustration of the mammalian TTFL circuit that generates the 24-hour rhythm. *Transcription factors Circadian Locomotor Output Cycles Kaput (CLOCK) and Brain and Muscle ARNT-like 1 (BMAL1 or ARNTL1) dimerise to activate the transcription of Period (PER, with paralogs Per1/2/3) and Cryptochrome (CRY, with paralogs CRY1/2), the protein products of which in turn inhibit CLOCK:BMAL1 and hence their own transcription, forming a negative feedback loop.*

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In these conventional TTFL models, it has **long been thought** that the rhythm generated by the core TTFL is transmitted downstream according to the central dogma (Fig XA). **Some of the core TTFL gene products directly or indirectly regulate the rhythmic transcription and degradation of other genes (mention Per??)**, leading to transcript-level oscillation, limiting translation rate, and causing protein-level oscillation as well. From this, protein activity rhythmicity arises, leading to cell functions **and physiological** rhythmicity (e.g. metabolism and signalling rhythms).

However, the **animal circadian model** has been challenged because little direct evidence has been found for the conventional model, and there is mounting evidence showing discrepancies in this model, which has been summarised previously (Mihut et al., 2025; Stangherlin et al., 2021). **Some counter arguments include the low amplitude in protein rhythms, which is much lower than that found in physiological outputs, the poor overlap between all rhythmic transcripts and rhythmic proteins, and the huge inconsistency in the phases of many transcripts and their**

Commented [EL3]: A major limitation of the TTFL model in its current state is its inability to provide a clear mechanistic explanation for how the robust, approx. 24 hour periodicity of the system arises. Negative gene expression feedback loops are common mechanisms of mRNA homeostasis that are ubiquitous in cell biology. They are often employed in the propagation and regulation of signal transduction networks, e.g. NF-kB (oscillates with a period of around 2–3 h) and ERK signalling pathways (approximately 3–4 h) [47,48]. These cycles usually take just a few hours to complete, reflecting the time required for transcription, RNA processing, translation and repression/inactivation to occur and rapidly damp back to steady state within a few cycles. ???

proteins, even if both of them do oscillate. This has driven the development of a post-transcriptional rhythmicity model, which suggests that the cellular timing mechanism directly generates rhythmicity in protein activity through post-transcriptional regulation (e.g. allosteric regulation, covalent modification, sequestration, **turnover** rate regulation), which then leads to rhythmic cell functions (James & O'Neill, 2025; Mihut et al., 2025). **Recently, evidence for greater circadian post-translational control has been presented for the plant system as well (Mehta et al., 2025).**

Yet this raises another question: if transcript rhythmicity is not the main driver of circadian rhythm, what then accounts for the strong transcript oscillation observed experimentally? One hypothesis is that this oscillation is actually an epiphenomenon of messenger RNA (mRNA) level homeostasis conferred by the TTFL output (Fig XB). Through balancing RNA synthesis and decay rates, the impact of systemic rhythmicity input (e.g. insulin, glucocorticoid signalling) can be buffered, and the mRNA levels can be maintained such that they do not become limiting for the translation of proteins encoded. **This is the main hypothesis upon which this study is built.**

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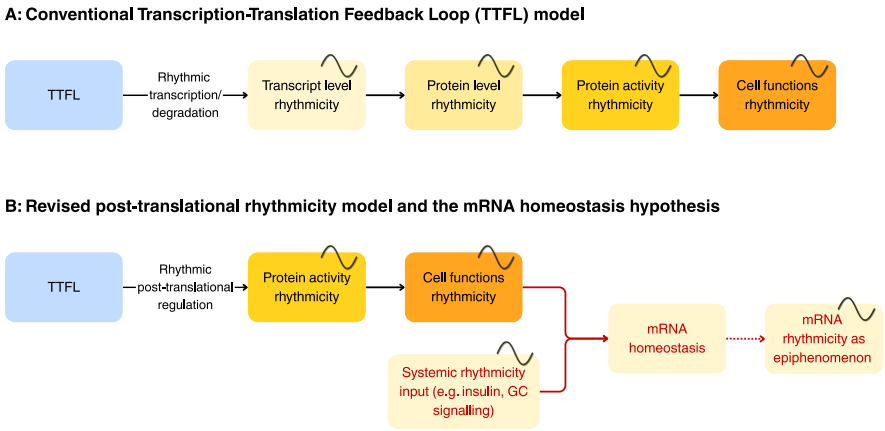

Fig. X: Schematic illustration of (A) the conventional TTFL model and (B) the revised post-translational rhythmicity model with the mRNA homeostasis. *The conventional model suggests that circadian information flows according to the central dogma. The revised post-translational*

regulation model suggests a direct flow of rhythmicity information from the TTFL to the rhythmicity of protein and cellular functions. Highlighted in red shows the hypothesis that mRNA rhythmicity is an epiphenomenon of mRNA homeostasis maintenance downstream of the TTFL output.

If this hypothesis is true, knocking out key components of this core TTFL (e.g., Per1, Bmal1) should disrupt homeostasis and lead to widespread **changes** in the transcript levels of the regulated genes. And if the daily rhythms seen in transcript level are indeed an epiphenomenon of homeostasis, then the rhythmic transcripts should be the ones most regulated, hence their levels are expected to change most in the knockout (KO). Therefore, a system-wide differential expression analysis of rhythmic genes in time-course transcriptomics data would be a good way to test this hypothesis.

Previous studies have addressed related questions using time-course microarray data from wild-type (WT) and Clock KO mice, where the majority of rhythmic genes (70-80%) were found to be differentially expressed using two-way ANOVA (Miller et al., 2007). **With modern RNA-seq technology, more comprehensive time-course RNA-seq data on** TTFL KO **organisms and modern differential expression analysis methods have become available.** Therefore, this study aims to systematically evaluate this hypothesis across multiple TTFL KO genotypes using published RNA-seq data and modern statistical transcriptomics analysis techniques.

Methods

MENTION PLOTS ARE MADE WITH LOG2 VALUES!

Acquisition of RNA-seq datasets

A strict criterion of "WT vs core circadian clock KO", "two or more replicates", "4 hr or shorter sampling interval", "24 hr or longer experiment duration" was used to select mouse and Arabidopsis time-course RNA-seq datasets.

For the mouse model system, raw FASTQ data derived from mouse kidney cells in WT condition and post-natal renal Bmal1 KO (Accession number: GSE216252, Bignon et al., 2023) was retrieved and raw gene-level count data derived from (1) mouse liver cells in WT condition and Per1/2 dKO (Accession number: GSE171975, Aviram et al., 2021), (2) mouse liver cells in WT condition and Cry1/2 dKO (Accession number: GSE135898, Weger et al., 2021), and (3) mouse liver cells in WT condition and post-natal whole organism Bmal1 KO (Accession number: GSE70499, Yang et al., 2016) were retrieved. All mouse datasets were downloaded from the Gene Expression Omnibus (GEO). For the Arabidopsis model system, RNA-seq datasets were retrieved from whole rosettes in WT, *lhy cca1*, *prr7 prr9* and *toc1* plants (Obtained from the supplementary data of the preprint by Mehta et al., 2025, as the raw data associated with accessions PRJEB103758, PRJEB103755, PRJEB103756, PRJEB103757, were not publicly available at the time of analysis). More details on the datasets are available in the **Supplementary Information**.

RNA-seq data processing

The renal Bmal1 KO FASTQ files were processed using the nf-core RNA-seq framework (Ewels et al., 2020) (v3.12.0) on Nextflow (Di Tommaso et al., 2017), with the murine genome GRCm39 and genome annotation vM31 from the Genome Reference Consortium. Specifically, low-quality 3' ends and adapters were trimmed and filtered for low complexity with fastp (S. Chen, 2025) (v0.23.4). Then the reads were aligned against the reference genome using STAR (Dobin et al., 2013) (v2.7.9a), and then summarised with Salmon (Patro et al., 2017) (v1.10.1) to yield the raw counts.

The raw counts generated from the RNA-seq QC or retrieved directly from the original **paper** was analysed with the pipeline. **First, as genome mapping of the datasets was performed differently**

across datasets, a common annotation genome was used for consistency (Ensembl Mus musculus genome release 102). The raw counts were filtered for protein-coding genes, then for genes with total counts per million (CPM) greater than or equal to 1 in at least 2 samples (or greater than or equal to 5 in at least 3 samples for the Per1/2 dKO dataset). Raw counts of the filtered genes were then normalised with Trimmed Mean of M-values (TMM) using edgeR (Y. Chen et al., 2025), converted to CPM values with a prior.count of 2 and a **$\log_2$** transformation was applied. The data was further processed to correct for unwanted variation with the remove unwanted variation (RUV) methods (Risso et al., 2014). Specifically, the RUVs method from the R Bioconductor package RUVseq was used. All genes were used as the negative control for RUVs, for RUVs' robustness towards control quality and their emphasis on replicate differences **reported by the original paper**. The number of factors of unwanted variation was set to two (k = 2) for RUVs to minimise the removal of biological signal. The RUV normalised matrix was used for rhythmicity analysis, while the two covariate matrices generated from RUVs (W1 and W2) were used for **linear model-type DE analysis** with the pre-RUV count matrix, **as illustrated in the original paper**.

Commented [EL5]: so dodgy bro..

PCA and RLE plots

Principal Component Analysis (PCA) was performed to assess sample clustering. The first two principal components were plotted for the filtered unnormalised, TMM normalised data, and TMM-RUV normalised data. Additionally, relative Log Expression (RLE) plots were also generated for these data to assess distributional consistency.

Rhythmicity detection

JTK_CYCLE (Hughes et al., 2010) was applied with the default settings to identify 24-hour period rhythms. For the Per1/2 dKO and Arabidopsis datasets, the uneven replicate counts due to **missing** samples were accounted for by defining a specific number of samples in each replicate group. For fixed threshold filtering, genes were classified as 'rhythmic' using the Benjamini-Hochberg (BH) adjusted p-value (p_{adj})**: 'rhythmic' if $p_{adj} < 0.05$ and relative $\log_2$ AMP > 0.29, 'non-rhythmic' if $p_{adj} > 0.1$ or relative $\log_2$ AMP < 0.29,** and the remaining genes were classified as 'ambiguous'.

DryR (Weger et al., 2021) was applied with the default setting, and genes with chosen_model_BICW < 0.65 were classified into the ambiguous group 0 (Bignon et al., 2023). Relative amplitude thresholding was also done with a threshold of 0.58.

Relative amplitudes were separately calculated using LimoRhyde (Singer & Hughey, 2019) instead of directly using the output of JTK_CYLCE. As JTK_CYCLE is a non-parametric, rank-based method, its amplitude estimates are not directly comparable across runs, making them less suitable for consistent thresholding. LimoRhyde was applied with a rhythmicity detection design matrix to yield the coefficients for the sine and cosine terms, which were used to calculate the relative amplitude of the $\log_2$-transformed data:

$$A = \sqrt{a^2 + b^2}$$

where A **was** the $\log_2$ relative amplitude and a and b were the coefficients. This relative amplitude was related to the linear absolute amplitude in the following way:

$$A_abs = 2^{\mu + A} - 2^{\mu}$$

where A_abs was the linear absolute amplitude and μ was the $\log_2$ mean.

Differential expression detection

Differential expression analysis was carried out with harmonic regression using LimoRhyde (Singer & Hughey, 2019), which is mainly based on Limma (Ritchie et al., 2015). To detect differential expression while accounting for differential rhythmicity, the LimoRhyde's differential rhythmicity design matrix, together with the two covariate terms from RUV, was adopted. Following the notation of the original authors, the linear model used for this study could be expressed as:

$$E(y_{gi}) = \beta_{g0} + \beta_{g1} h_i + \beta_{g2} \cos\theta_i + \beta_{g3} \sin\theta_i + \beta_{g4} h_i \cos\theta_i + \beta_{g5} h_i \sin\theta_i + W_1 + W_2$$

where $E(y_{gi})$ is the $\log_2$ expected expression level for gene g in sample i, θ_i is the circadian time in radians, β_{gj} are the coefficients, h_i is the binary genotype variable (0 for WT, 1 for mutant) and W_1 and W_2 are the covariates from RUV. Note that the relatively high number of terms is acceptable for harmonic regression, given the sufficient total sample size.

The genotype term was assessed using the function topTable to obtain the BH-adjusted p-value (p_{adj}) and $\log_2$FC for thresholding and classification of DE**: 'up-regulated' if $\mathbf{p_{adj} < 0.05}$ and $\mathbf{\log_2 FC > 0.58}$, 'down-regulated' if $p_{adj} < 0.05$ and $\mathbf{\log_2 FC < -0.58}$, and the remaining genes were classified as 'unchanged'.**

Gene Set Enrichment Analysis

GO term enrichment tests on the Biological Process (BP) were performed using the Bioconductor ClusterProfiler package (Xu et al., 2024). The **Ensembl Mus musculus genome release 102** was used as the background reference. Significance was determined with default settings, with the BH-adjusted p-value threshold of 0.05.

Results

Overview of the rhythmicity and differential expression detection pipeline

To test whether breaking the circadian clock leads to transcriptome-wide changes in the levels of rhythmic genes, various time-course RNA-seq datasets from wild-type and clock gene mutants of Mus musculus and Arabidopsis thaliana were analysed and compared using the rhythmic differential expression pipeline developed (*Fig X*). For the complete pipeline, raw FastQ files from RNA-seq are used to generate a raw count matrix, which is then normalised and analysed in parallel for rhythmicity and differential expression. The results are combined to yield a list of WT rhythmic genes that are differentially expressed, which could be further analysed.

Commented [EL6]: or need to specify completely arrythmic?

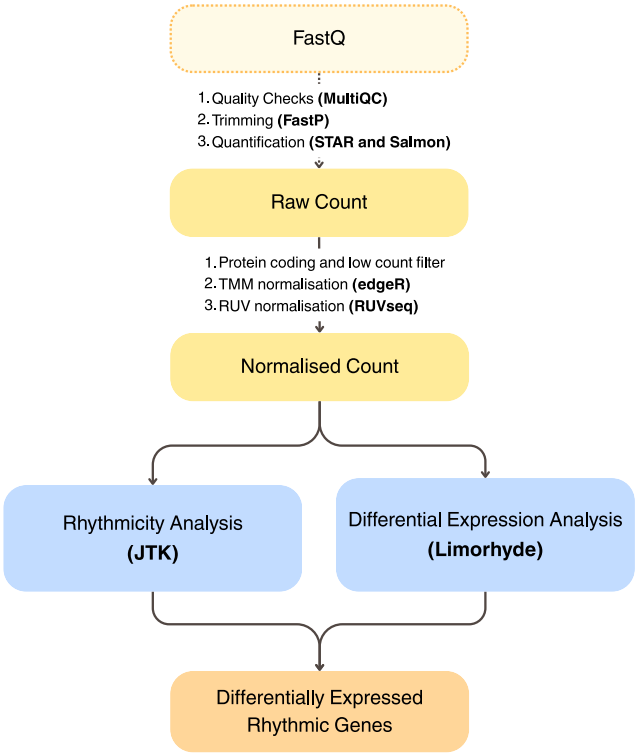

Fig X: Schematic illustration of the rhythmic differential expression analysis pipeline. *For the complete pipeline, FastQ files from RNA-seq experiments are processed, mapped to the reference genome, and quantified to produce the raw count matrix. This matrix is then normalised and subjected to rhythmicity and differential expression analysis to identify the differentially expressed rhythmic genes. The specific methods used for each step are listed in brackets. The QC step is shown in dotted lines, as only one dataset underwent it.*

Validation and refinement of the pipeline

Mouse WT and renal Bmal1 mutant RNA-seq data were selected as the starting dataset for building the pipeline, given their high replicate number (n = 5), which provides high statistical power, **and the comparable normalisation pipeline in the original paper**. The validated pipeline was then applied to other datasets with as much consistency as possible.

Raw FastQ files from RNA-seq experiments were processed to yield the raw count matrix. **Before normalisation, the raw count was filtered for protein-coding genes to remove noise from non-coding genes.** Normalisation of the raw count was carried out in two steps: (1) TMM normalisation to correct for library size differences and (2) Remove Unwanted Variation (RUV) normalisation to remove hidden technical variations (e.g. batch and other technical effects) across samples (Material and Methods). This method was consistent with that used in Bignon et al. 2023.

That RUV removed unwanted variation without removing **much** meaningful biological structure was carefully validated with the oscillation profile of genes before and after RUV **(include figure?),** relative log expression (RLE) (Fig X), and **principal component analysis (PCA) plots (Fig X).** The oscillation profile of example genes selected showed that RUV cleaned up the data and smoothed out the sinusoidal temporal behaviour for some genes. The RLE plots further showed the effect of RUV in levelling the sample variances. In addition, PCA revealed that RUV successfully separated the samples by circadian phase, thereby unravelling biologically meaningful information from noise.

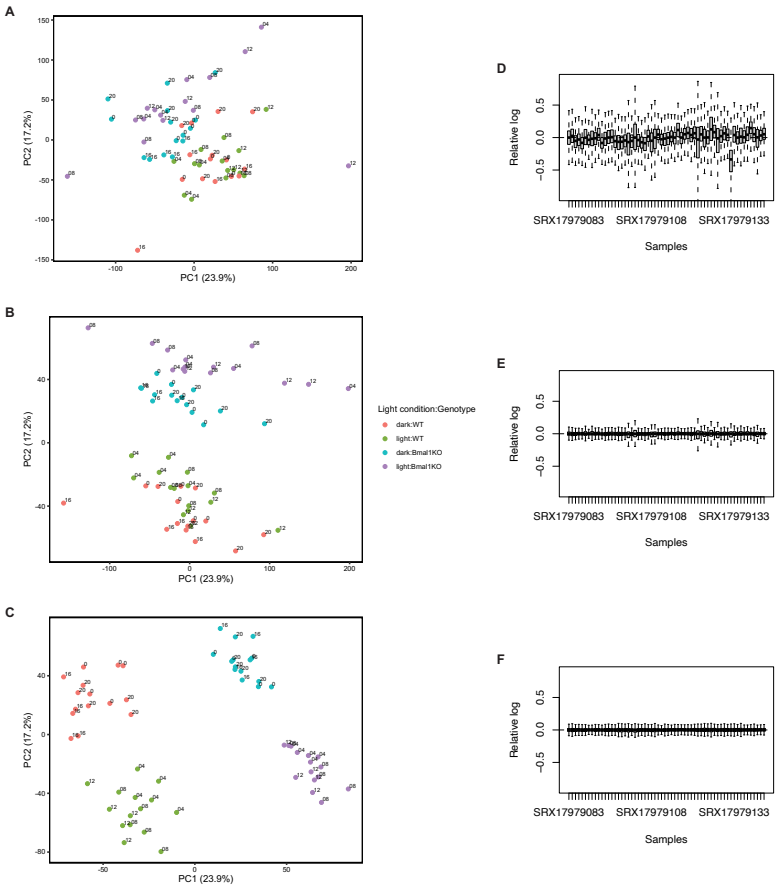

Fig X: PCA and RLE plots showing that RUV reduces unwanted variation across samples for the Renal Bmal1 KO dataset. *(A-C) PCA plots of the filtered unormalised (A), TMM normalised (B), and TMM-RUV normalised (C) data, coloured by light condition and genotype. (D-F) RLE plots showing relative* log_2 *expression of all samples in the filtered unormalised (D), TMM normalised (E), and TMM-RUV normalised (F) data.*

To identify genes that are rhythmic in wild type, rhythmicity analysis was carried out with JTK_CYCLE, a widely used non-parametric method, for its stringency and conservativeness to

minimise false positives. Both DryR and JTK_CYCLE were considered originally. Without ambiguity and amplitude thresholding, a good overlap was seen (Fig XA) for the number of rhythmic genes. The number is slightly higher than that reported in literature for DD (Otobe et al., 2026; Zhang et al., 2014), which is reasonable that light input induced more rhythmicity. However, under our stringent $\log_2$ relative amplitude (rAMP) threshold (**adjusted p-value < 0.05, rAMP > 0.29, justified below**) and ambiguity binning (Materials and Methods), there was limited overlap in high confidence rhythmic genes detected by the two methods (Fig X). This was likely due to the different aim and rhythmicity detection rationale (e.g. symmetry assumption?) of the two methods - dryR was built to classify differential rhythmicity modes between different conditions instead of directly probing for rhythmic genes in WT. Hence, JTK_CYCLE was deemed more suitable for our study. This also suggested that the subset of high-confidence rhythmic genes strongly depends on the analytical method. For comparison, LimoRhyde's rhythmicity detection method, which was parametric and regression-based, was also attempted, and it showed good overlap in high-confidence rhythmic genes (Fig X).

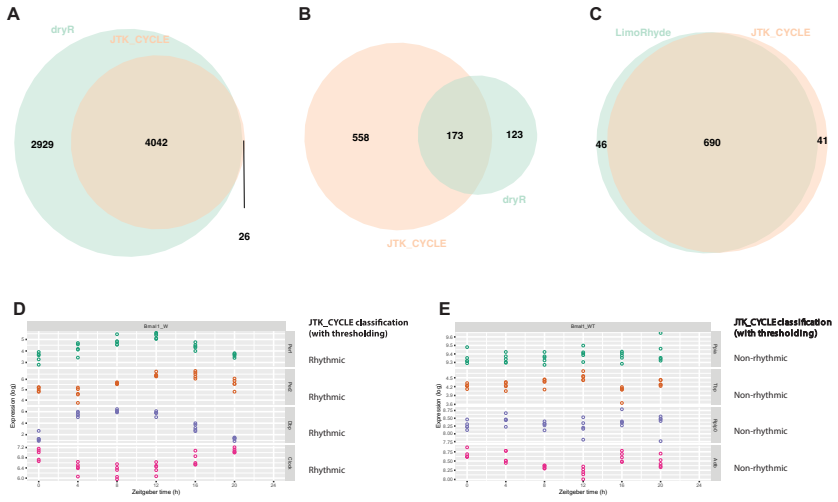

Fig X: Rhythmicity detection method validated using the Renal Bmal1KO dataset. *(A-B) Comparison between the rhythmic genes identified by dryR and JTK_CYCLE. A: no ambiguity and*

amplitude filter; B: dryR results filtered with BICW<0.65 and rAMP<0.29, JTK_CYCLE results filtered with rAMP>0.29). rAMP represents the log_2 relative amplitude. (C) Comparison between the rhythmic genes identified by LimoRhyde and JTK_CYCLE. (D-E) Time-course plots of a few known rhythmic (Per1, Per2, Dbp, Clock) and housekeeping, non-rhythmic (Ppia, Tbp, Tplp0, Actb) genes and their corresponding rhythmicity classification by JTK_CYCLE with amplitude thresholding.

Differential expression detection was carried out using LimoRhyde. For time-course differential expression analysis with rhythmicity, simple mean comparison (e.g. t-test) was not appropriate due to the inability to distinguish between sample variance and amplitude. This was account for by LimoRhyde with cosinor regression – the inclusion of sine/cosine terms in the design matrix. The use of limma-trend over voom on this dataset as suggested by LimoRhyde was justified by the consistent sample library size (Coefficient of Variation = 5.31%), which allows the use of this simpler and more robust method to handle the mean-variance relationship. The mean-variance plot (Fig X) was also inspected to adjust the low count filtering threshold (Materials and Methods) and ensure limma-trend successfully captured the mean-variance trend.

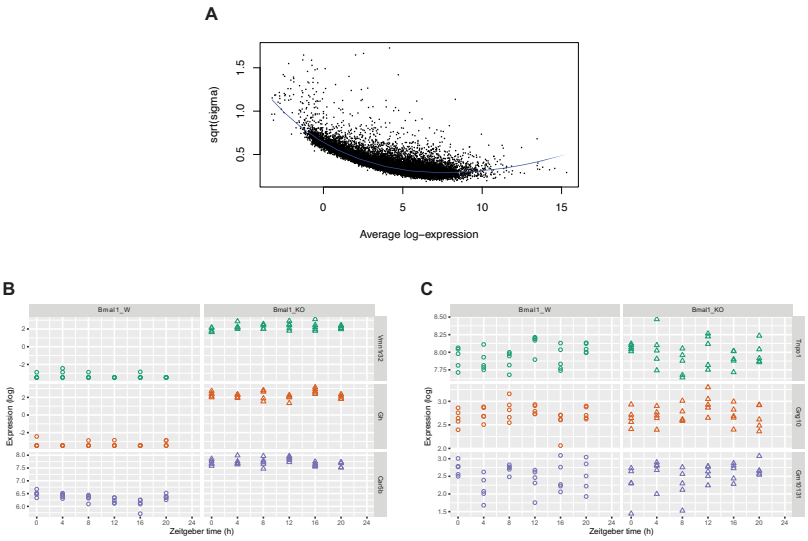

Fig X: Differential expression analysis method validated using the Renal Bmal1KO dataset. *(A) Log residual standard deviation versus average log expression plot showing the mean-variance relationship of the fitted LimoRhyde linear model. (B-C) Time-course plots of the most and least differentially expressed genes assessed by LimoRhyde.*

The rhythmicity and differential expression detection pipelines were validated with multiple time-course oscillation plots. A few known rhythmic genes (Per1, Per2, Dbp, Clock) and non-rhythmic, housekeeping genes (Ppia, Tbp, Rplp0, Actb) were plotted and compared with the JTK classification result. Under stringent rhythmicity threshold, the classification was mostly consistent with expectation, with Clock and Ppia classified as 'Ambiguous'. **As only genes with strong and highly confident oscillation were classified as 'Rhythmic' to reduce false positives, this was deemed a reasonable output of rhythmicity analysis.** For differential expression validation, the three genes with the highest adjusted p-values in LimoRhyde were plotted, showing a large change in mean level and a small variance. The opposite was observed for the lowest three.

Commented [EL7]: expected DE & non-DE validation? but don't think we know lol

Datasets Selection

To understand whether a dysfunctional circadian clock led to system-wide differential expression of rhythmic genes, time-course RNA-seq experiments of WT and clock mutants with sufficient replicates ($n \geq 2$) and timepoints (at least 6 timepoints in more than 24 hours) were searched for in the literature. Four **mouse** datasets of three types of clock KO (Bmal1, Per1/2 and Cry1/2) were selected for analysis using the pipeline described above. Note that whole organism pre-natal Bmal1 knockout datasets (e.g. Atger et al., 2015) were deliberately excluded due to the severe phenotypic consequences caused by loss of this key transcription factor (Yang et al., 2016), which would confound the interpretation and conclusion.

For the selected datasets, due to different research question asked during the experiment and different experimental setting (e.g. mouse strain, light/feeding condition, organ sampled, replicate number, experimental duration, timepoint size, gene KO method and RNA-seq method), they each have their own limitations (summarised in Table X, **detailed dataset information in Supplementary Information**) and careful selection and interpretation was required.

Commented [EL8]: elaboration on criteria??

In **Bignon's experiment, only the kidney cells possessed the Bmal1 KO and systemic circadian influence was reported (Bignon et al., 2023). In Aviram's experiment, the mice were on a free clock without external light input (DD), making it less comparable to other datasets (LD).** Also, UMI-based RNA-seq method was used to generate the raw count in this dataset, leading to a different mean-variance profile from other datasets generated with the standard RNA-seq method and requires pipeline adaptations, as explained later. The Weger's dataset had only two replicates and hence a lower, marginal statistical power. The Yang's dataset did not seem to show complete Bmal1 knockout penetrance in the liver (Yang et al., 2016), and hence later deemed unsuitable for our pursuit.

Dataset	Mutation	Tissue	Light condition	RNA-seq method	KO method	Replicate number
Bignon et al.	Bmal1 (Renal)	Kidney	LD	Standard	Complete penetrance??	5
Aviram et al.	Per1 Per2 (Whole)	Liver	**DD**	**UMI-based MARSeq**	Complete penetrance??	3
Weger et al.	Cry1 Cry2 (Whole)	Liver	LD	Standard	Complete penetrance??	**2**
Yang et al	Bmal1 (Whole, post-natal)	Liver	**DD**	Standard	**Incomplete penetrance**	3

Table X: Summary of the mouse datasets analysed and their limitations. *Detailed information about each datasets could be found in the Supplementary Information.*

Despite the differences, a unifying trend was observed in (1) the absence of transcriptome-wide expression level change and (2) the stronger differential expression in high amplitude rhythmic genes, which will be discussed in detail later.

Unfortunately, for Arabidopsis, no datasets with fully dysfunctional clock mutants possess enough replicates required for **DE** and rhythmicity analysis. However, a high-quality dataset (Mehta et al., 2025) of a few photoperiod mutants (**WT, CCA1/LHY, PRR9/PRR7, and TOC1,** functional clock but altered period) was found and analysed, which would serve as a good control for the complete clock KO datasets if available in the future.

Renal Bmal1, Per1/2 dKO, Cry1/2 dKO datasets selected and validated for analysis

To validate that the circadian clock in the selected datasets was indeed dysfunctional, (1) time-course plots (Fig X) of a few core clock genes (Per1/2/2, Cry1/2, Arntl **or Bmal1**, Clock, Nr1d1, Dbp) and known non-oscillatory housekeeping genes (Actb, Rplp0) and (2) Venn diagrams of rhythmic genes in WT and mutant (Fig X) were generated (Fig X, or supp?).

Firstly, the datasets were inspected for complete and successful clock protein KO. In the time-course plots, a surprisingly high transcript level was observed for the gene knocked out in all datasets. This could be explained by the KO technique used in these experiments – instead of a complete gene excision, only certain exons were excised to make the protein unfunctional. The partial transcripts were detected in RNA-seq and subsequent quantification, leading to the high level of KO gene in the mutant. The successful KO for the renal Bmal1, Per1/2 and Cry1/2 datasets were validated by rt-qPCR and Southern blot (Horst et al., 1999; Nikolaeva et al., 2016; Zheng et al., 2001). For the whole-organism post-natal Bmal1 KO, incomplete penetrance was observed in the mouse liver which was the tissue sampled for RNA-seq (Yang et al., 2016) and some analysed retain a functional circadian clock. This explains the higher variance among the sample seen in the mutant time-course plots. As such, this dataset was not considered valid for drawing conclusions for our hypothesis.

Commented [EL9]: also messy PCA?

Secondly, the functional depletion of the clock was checked. In the Per1/2 and Cry1/2 time-course plots, all the clock genes showed a clear loss of rhythmicity, suggesting a dysfunctional clock. In contrast, the renal Bmal1 KO dataset showed oscillation in Per1/2/3, Clock, Nr1d1 and Dbp in the time-course plots (Fig X) and a significant number of genes remained rhythmic in the mutant (Fig XA). This was an expected observation in kidney-targeted Bmal1 KO – the organism-level

circadian clock remained intact and exerted systemic influence on various genes in the kidney cells, making them rhythmic despite the absence of functional BMAL1. As BMAL1 is a key driver in **mammalian** clock feedback network, the absence of which theoretically should lead to a completely dysfunctional clock. This is further supported by the complete loss of rhythmicity of Cry1/2 and Npas2. Therefore, the renal Bmal1, Per1/2 and Cry1/2 datasets were deemed suitable for our hypothesis.

Commented [EL10]: possibility of Per/Cry oscillation maintaining transcriptome level?

Fig X: Comparison between rhythmic genes in WT and clock gene KO in the three datasets. *Obtained from JTK_CYCLE under the fixed threshold.*

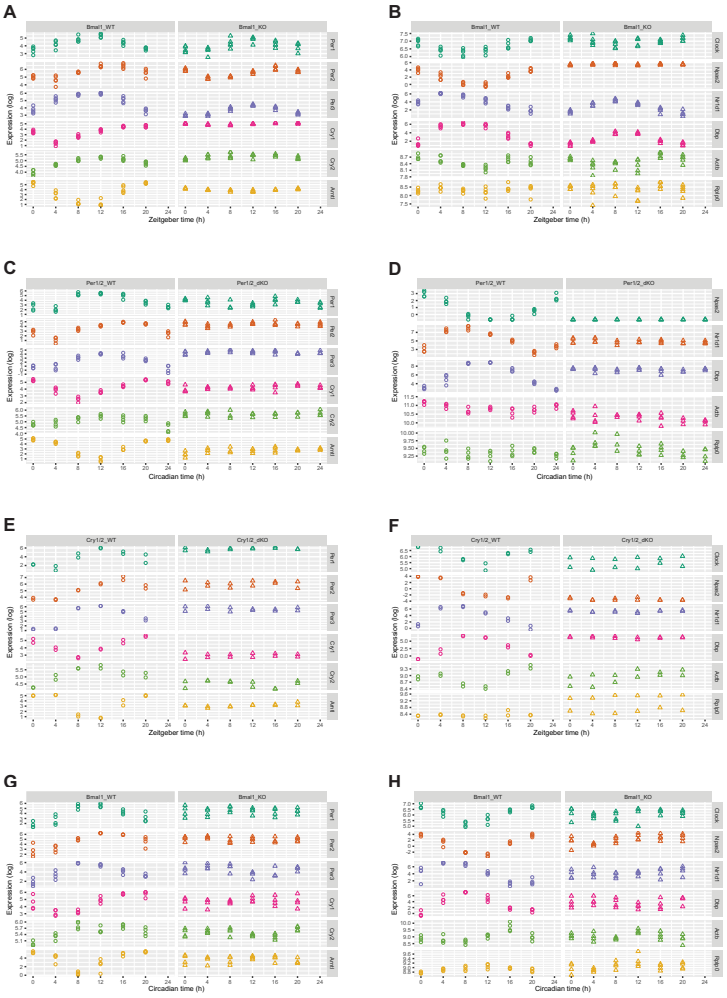

Fig X: Time-course plots of some known rhythmic and non-rhythmic genes for each dataset. *(A-B) Renal Bmal1 KO dataset. (C-D) Per1/2 dKO dataset. (E-F) Cry1/2 dKO dataset. (G-H) Whole organism post-natal Bmal1 KO dataset. Rhythmic genes: Per1, Per2, Per3, Cry1, Cry2, Arntl (Bmal1), Clock, Npas2, Nr1d1, Dbp; Non-rhythmic genes: Actb, Rplp0.*

Before applying LimoRhyde to the datasets, the residual standard deviation versus average log expression plots were reviewed to ensure the datasets have a reasonable mean-variance relationship under the current low-count filter. The Bmal1 and Cry1/2 KO data showed expected variance reduction with increasing mean expression and the trend was well predicted with the blue curve fitted by limma-trend. In contrast, the Per1/2 data showed significant damping at low count. This was a characteristic pattern for UMI-based RNA-seq data due to the collapse of transcript reads with identical UMI. To remove the bias generated by the low variance at low count, stricter low count filtering was applied on the Per1/2 datasets, resulting in a more appropriate mean-variance relationship for downstream analysis. As a routine, the PCA plots (Fig. X) were inspected too, like the renal Bmal1 data, the Per1/2 and Cry1/2 data showed gained circadian phase clustering after TMM and RUV normalisation.

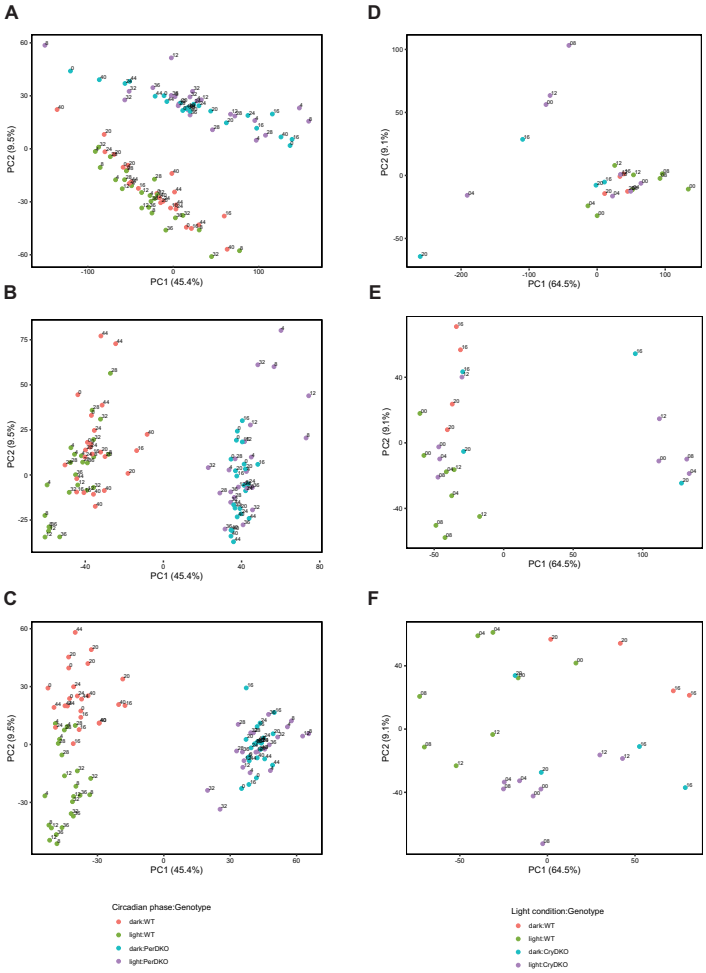

Fig X: PCA plots showing that RUV reduces unwanted variation across samples in the Per1/2 dKO and Cry1/2 dKO datasets. *(A-C) PCA plots of the filtered unormalised (A), TMM normalised (B), and TMM-RUV normalised (C) Per1/2 dKO data, coloured by light condition and genotype. (D-F) The same for Cry1/2 dKO.*

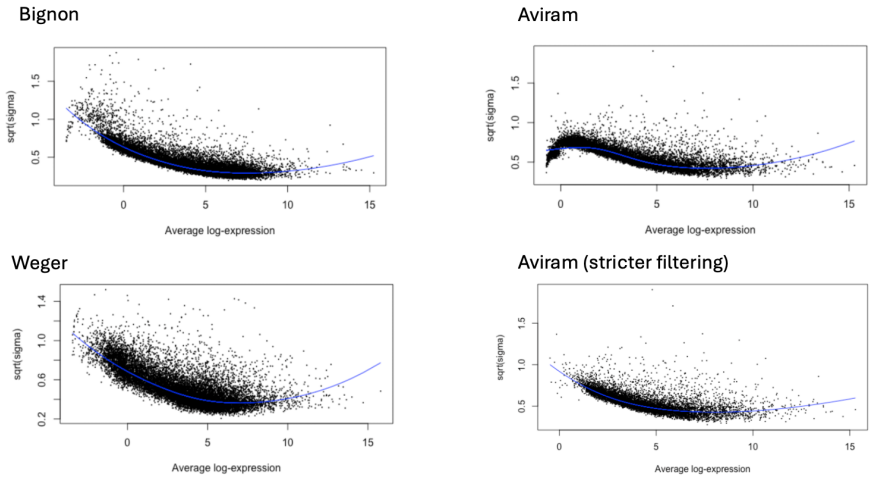

Fig X: Extra filtering was needed for the Per1/2 dKO dataset. *Log residual standard deviation versus average log expression plot showing the mean-variance relationship of the fitted LimoRhyde linear model. (A) Bmal1 KO; (B) Cry1/2 dKO; (C,D) Per1/2 dKO before and after more stringent filtering.*

Volcano plots were generated to visualise differential expression. It was noticed that the adjusted p-values in the Cry1/2 dataset is generally bigger than the other two datasets, giving it a flattened volcano plot. This could be explained by the low replicate number (n=2) of the Weger et al. dataset, which reduced the confidence for each DE gene.

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DE-rhythmicity threshold analysis reveals a relationship between amplitude and DE

Running each selected datasets through the rhythmicity and DE detection pipeline, the rhythmicity adjusted p-value, log2 relative amplitude, DE adjusted p-value and log2 fold change were calculated for each gene. Using selected adjusted p-values (0.05 for all our analysis), rhythmicity and DE threshold, genes could be assigned one of rhythmic, non-rhythmic, ambiguous and one of up, down and unchanged. And the proportion of DE and rhythmic genes out of all rhythmic genes could be obtained, which is simply referred to as DE-rhythmic ratio below.

Interestingly, increasing the $\log_2$ relative amplitude threshold gave an increase in the DE-rhythmic ratio and is true with different logFC thresholds in all three datasets, shown by the plot of DE-rhythmic ratio against relative amplitude plot (Fig X). **Note that the $\log_2$ relative amplitude greater than 1 was possible due to the normalisation steps which renders some transcript levels negative.** For relative amplitude threshold above 1, the number of DE-rhythmic genes became too small (fewer than 10) in the Bmal1 and Per1/2 datasets, making the ratio unmeaningful for interpretation. However, for amplitude threshold below 1, a clear increasing trend could be observed, especially for the Bmal1 dataset (Fig XA) and this trend dampened with higher logFC threshold. This is shown more directly with the 3D plot of DE-rhythmic ratio against both logFC and amplitude thresholds (Fig X) – there was an increasing trend with amplitude threshold that dampens with increasing logFC threshold. This suggested that rhythmic genes that are differentially expressed with a dysfunctional circadian clock tend to have high amplitudes.

$$\text{DE-rhythmic ratio} = \frac{\text{\# DE \& rhythmic genes}}{\text{\# rhythmic genes}}$$

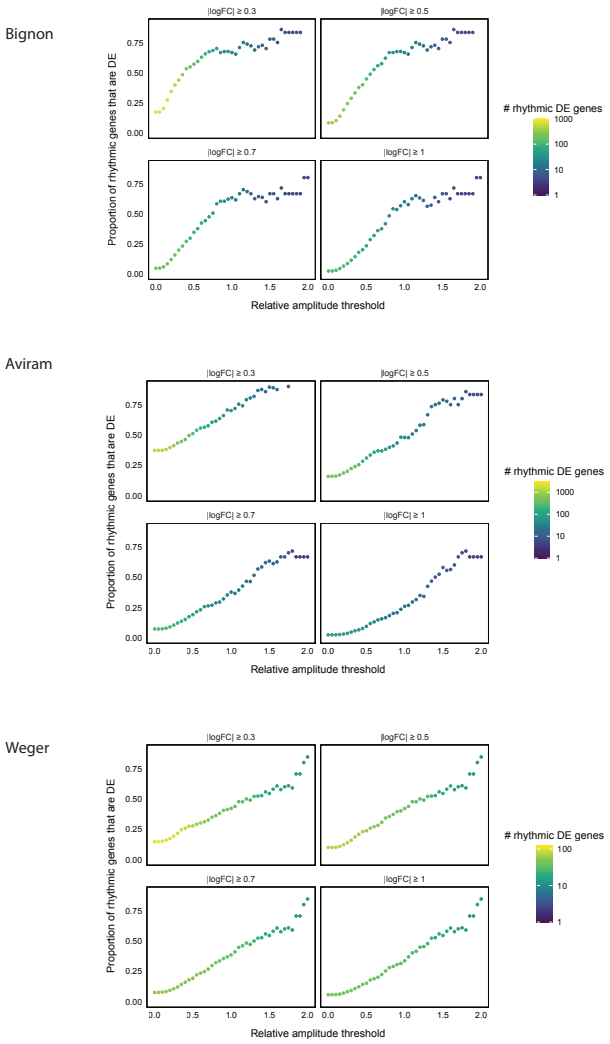

Fig X: Increasing relative amplitude threshold leads to higher DE-rhythmic ratio across various logFC thresholds. *Plot of DE-rhythmic ratio (proportion of rhythmic genes that are DE) against log_2 relative amplitude threshold for four different logFC thresholds (0.3, 0.5, 0.7 and 1). Data*

points coloured according the actual number of DE-rhythmic genes. A: Renal Bmal1 KO; B: Per1/2 KO; C: Cry1/2 KO.

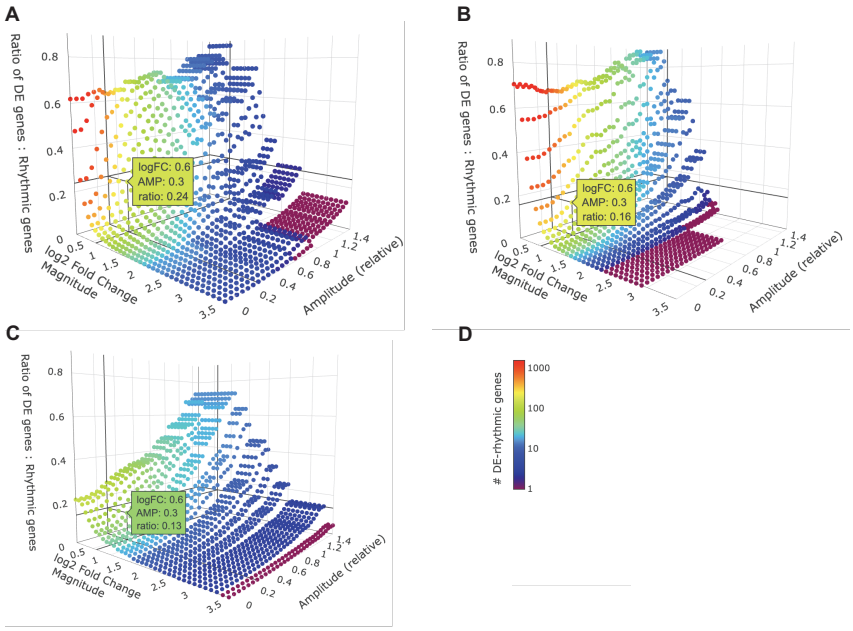

Fig X: 3D plots show a consistent positive relationship between amplitude threshold and DE-rhythmic threshold. *(A) Renal Bmal1 KO; (B) Per1/2 dKO; (C) Cry1/2 dKO; (D) Colorbar.*

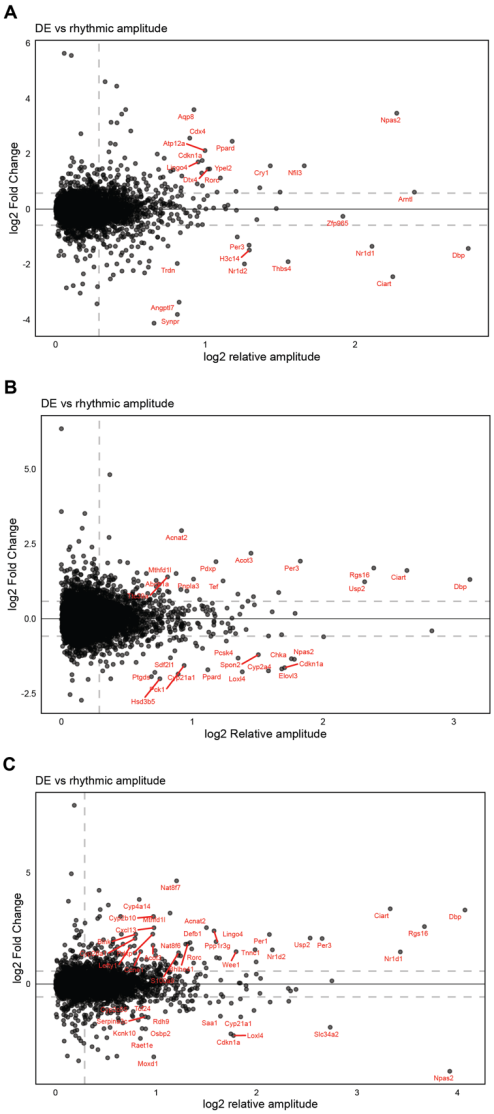

Fig X: LogFC vs relative amplitude plot with high value genes labelled.

Fixed threshold comparison of DE-rhythmicity disproves the clock-homeostasis hypothesis

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To classify rhythmicity and differential expression in a biologically meaningful way, thresholding on relative amplitude and $\log_2$FC magnitude **was** required in addition to significance-level thresholding. A widely used threshold to define an effective change in transcript level for circadian-related studies **was** at least a 1.5-fold change. Translating this into fold change yielded a $\log_2$FC of **0.58,** and for amplitude, a 1.5-fold peak-to-trough ratio corresponds to a $\log_2$ relative amplitude threshold of **0.29.** It was acknowledged that different datasets were obtained using different methods, hence their transcript levels and changes in levels are not directly comparable. Acknowledging this limitation while balancing practicality, a common threshold was used across datasets and **general trends, rather than absolute comparisons in DE-rhythmicity**, were explored.

Applying this threshold to all datasets, the DE ($p < 0.05$ and $|\log_2FC| > 0.58$) ratio for rhythmic ($p < 0.05$ and $\log_2rAMP > 0.29$), ambiguous, and non-rhythmic ($p > 0.1$ or $\log_2rAMP < 0.29$) genes could be calculated, as listed in Table X. Three interesting observations could be made. Firstly, under the current threshold, system-wide differential expression of rhythmic genes was unseen, with the DE ratio ranging from **13% to 34%**. This was directly against our hypothesis that the core clock is responsible for transcript-level maintenance. Secondly, despite the lack of system-wide DE expected, a **significantly** larger proportion of rhythmic genes are DE **(quote chi-square test here????).** Thirdly, the overall DE ratio was low, around **1% to 3%,** suggesting that the transcriptome **was** rather maintained when the circadian clock was dysfunctional.

A

Rhythmicity / DE	Non-rhythmic	Ambiguous	Rhythmic	Overall
Down	176	7	76	259
Unchanged	12470	67	557	13094
Up	130	8	98	236
% DE	**2.4**	**18.3**	**23.8**	**3.6**

B

Rhythmicity / DE	Non-rhythmic	Ambiguous	Rhythmic	Overall
Down	379	2	114	495
Unchanged	6520	45	1094	7659
Up	411	7	113	531
% DE	**10.8**	**16.7**	**17.2**	**11.8**

C

Rhythmicity / DE	Non-rhythmic	Ambiguous	Rhythmic	Overall
Down	119	17	18	154
Unchanged	11049	571	354	11974
Up	153	34	36	223
% DE	**2.4**	**8.2**	**13.2**	**3.1**

Table X: DE-rhythmicity results with fixed log_2FC and relative amplitude threshold. *Up: $p < 0.05$ and $log_2FC > 0.58$; Down: $p < 0.05$ and $log_2FC < -0.58$; rhythmic: $p < 0.05$ and $log_2rAMP > 0.29$; non-rhythmic: $p > 0.1$ or $log_2rAMP < 0.29$; ambiguous: remaining.*

Gene Ontology (GO) enrichment analysis suggests a possible explanation for amplitude-DE relationship

GO enrichment analysis was carried out on 1) genes that are both DE and rhythmic, 2) genes that are DE, and 3) genes that are rhythmic in WT, as defined by the fixed threshold, for each dataset (Fig X). Generally, the DE genes spanned a broad range of metabolic processes, and rhythmic genes were enriched in circadian-related processes and some metabolic processes. DE-rhythmic genes, as a subset of rhythmic genes, were similarly enriched in circadian-related processes, even more than rhythmic genes in the Per1/2dKO dataset. However, the DE-rhythmic results should be interpreted with caution, as the analysis was based on only **30-40** genes, which lies at the lower limit for robust statistical enrichment analysis. Although the GO analysis results are somewhat ambiguous, they suggest a possible hypothesis regarding the relationship between the DE-rhythmic ratio and the amplitude threshold: high-amplitude genes might be more directly linked to the TTFL, making their expression more susceptible to drastic changes following direct regulatory network disruption in the KO.

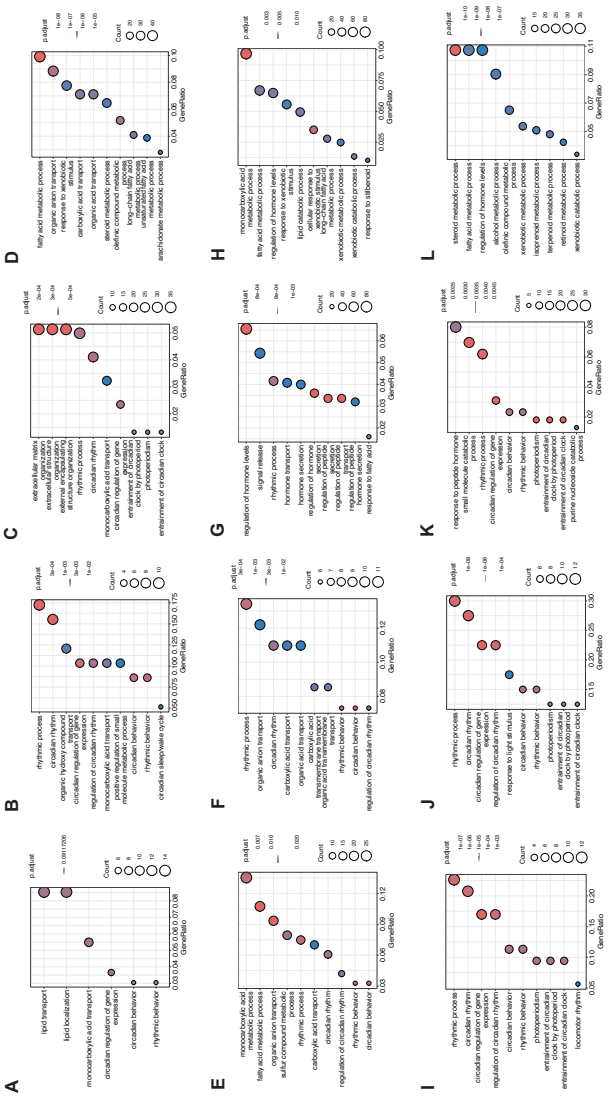

Fig X: GO enrichment analysis for Biological Processes showed strong enrichment in circadian-related processes for high amplitude genes. *(A-D) Renal Bmal1 KO; (E-H) Per1/2 dKO; (I-L) Cry1/2 dKO.*

Discussion

To test if system-wide DE of rhythmic genes could be observed for in clock KO mutants, a pipeline integrating time-course RNA-seq data analysis, rhythmicity detection (JTK_CYCLE) and DE analysis (LimoRhyde) was built and validated. Three mouse datasets (post-natal renal Bmal1 KO, Per1/2 dKO, Cry1/2 dKO) were for selected for our main investigation.

Biological and technical differences between datasets require caution during interpretation

As previously mentioned, drawing conclusions across datasets requires caution because of differences in experimental settings and data generation techniques, which can introduce significant variation in the transcriptome. For instance, mouse samples harvested during LD and DD should have different transcriptome as some genes are directly regulated by light input in LD but become non-rhythmic in DD. The feeding pattern (ad lib or restricted feeding) affects the metabolism of the mice. The organ harvested, replicate number, duration of entrainment / constant darkness before harvesting and even the mouse sex, age and strain might also affect the transcriptome. Strikingly, the technical differences (e.g. sequencing techniques) across different time-course RNA-seq experiments studies was reported to dominate over than biological variability and substantial inconsistencies were found even with highly consistent experimental designs (Brooks et al., 2023). Therefore, direct quantitative comparison between datasets was avoided and only qualitative comparison was performed with caution. And limitations exist with the application of a common amplitude and fold change threshold.

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Besides dataset variability, ambiguity in rhythmicity detection also posed a challenge. Different rhythmicity detection algorithms **(e.g. JTK_CYCLE, ARSER, eJTK_CYCLE, MetaCycle, dryR, LimoRhyde, LS, GeneCycle)** have been developed to identify periodic patterns in the transcriptome or proteome. However, they are based on different symmetry assumptions, rhythmicity definition, phase/amplitude calculation method and thresholding and hence give slightly different number of rhythmic transcripts and incomplete overlap (Miao et al., 2024). Although it was found that different methods agree on large number but the order of significance differ (Laloum & Robinson-Rechavi, 2020), which is consistent with our finding that dryR and JTK_CYCLE showed poor overlap with stringent rhythmicity thresholds. Also, it was reported that strongly rhythmic genes are more consistently detected by different methods and are **likely**

more biologically relevant (Laloum & Robinson-Rechavi, 2020). Therefore, stringency was prioritized over sensitivity for this study to minimise false positives, which is crucial for testing our hypothesis. JTK_CYCLE was selected for its robustness, conservativeness and non-parametric nature (i.e. more flexible waveform) and a amplitude threshold was applied to identify the group of high-confidence rhythmic genes.

As such, direct quantitative comparison between datasets was avoided. However, some common qualitative trends could be seen.

Lack of system-wide transcript level disproves general mRNA homeostasis hypothesis??

First and foremost, system-wide transcript-level dysregulation in rhythmic genes was not seen across all three mouse datasets (Table X). As explained in the Introduction, if the TTFL is responsible for maintaining mRNA homeostasis, widespread differential expression should be seen in rhythmic genes in the TTFL mutants. However, under our DE and rhythmicity thresholds, only 13% to 24% of all rhythmic genes showed significant changes in expression levels, and the overall DE rate was generally low (3% to 12%).

This **was** inconsistent with the findings of the previous analysis on the microarray dataset from Clock KO, which suggested a 70%-80% DE for the rhythmic genes. This might be due to that (1) microarray **captures a less complete transcriptome**, (2) their use of two-way ANOVA for DE analysis did not account for the variation generated by the rhythmicity, and (3) no amplitude and fold-change threshold were applied. Comparing with more recent RNA-seq analysis, this result **was** consistent with the that in the LimoRhyde's original paper (Singer & Hughey, 2019) which analysed whole organism pre-natal Bmal1 dataset excluded in our study (Atger et al., 2015). Only 237 out of 2434 rhythmic genes were found to be DE (adjusted p-value threshold for rhythmicity and DE: $p_{adj_rhy} \leq 0.01$ and $p_{adj_DE} \leq 0.1$) using LimoRhyde for both rhythmicity and DE detection. A similar analysis was performed in the paper that generated the renal Bmal1 dataset using dryR and harmonic regression, which found a DE ratio of 50% to 80% in the differential rhythmicity models with rhythmic WT (Bignon et al., 2023). This inconsistency with our finding using the same data might be due to the lack of any amplitude and FC thresholding in the paper, which classified nearly a third of all genes to be rhythmic and half to be DE. **As FC thresholding gives**

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more biological relevance and is a standard practice in differential expression analysis, one can argue that our conclusion that transcript level dysregulation was absent is reasonable. However, an interesting observation was that the DE-rhythmic ratio was still significantly higher for rhythmic genes and echoed with our observations in amplitude-DE trend.

Commented [EL15]: is our DE threshold too strict lolll, a bit absurd

Strongly rhythmic genes tend to be more DE???

Varying the amplitude and $\log_2$FC threshold gave an interesting observation of a consistent increase in DE-rhythmic ratio at higher amplitude threshold with various fixed $\log_2$FC threshold. This suggests that gens with high amplitude tend to be more differentially expressed in the KO. This also echoed the higher DE-rhythmic ratio for rhythmic genes we saw with a fixed threshold earlier. One possible explanation was that the mRNA homeostasis hypothesis is partially true – only a group of strongly rhythmic genes with high amplitudes were under level maintenance by the TTFL downstream output.

However, a more straight-forward explanation would be that these strongly rhythmic genes shows strong oscillation as they are closely related to the core TTFL that generates the daily rhythm. Breaking the core TTFL, therefore, would logically disrupt the circadian-related genetic networks and lead to DE of the genes involved. This was supported by the observation that most DE genes with the highest amplitudes in the three datasets were circadian-related (e.g. Ciart, Dbp, Rgs16, Npas2) and GO enrichment analysis also showed that DE-rhythmic genes were highly enriched in circadian-related biological processes rhythmic genes, similar or even more than that for all rhythmic genes. **Putative gene regulatory network (GRN) analysis** of the DE-rhythmic genes might be able to provide insight into whether these genes are enriched for shared regulatory connections. If yes, DE in high amplitude genes would be a direct regulatory consequence of knocking out key components of the TTFL.

Alternative explanations for transcript rhythmicity

According to the post-translational rhythmicity model, transcript rhythmicity is not the cause cell functions rhythmicity. Yet the reaon behind transcript oscillation is unclear. Our analysis and results disfavour the mRNA homeostasis hypothesis in the mammalian system. Alternatively, the

rhythms in transcripts might be a consequence of daily changes in chromatin accessibility caused by protein activity rhythmicity.

…

Limitations and further work

As mentioned earlier, the biological and technical differences between datasets limited the extend of comparison between different datasets. However, unless large-scale, time-course experiments on multiple clock KO are done in parallel, it is practically difficult to circumvent this problem. **Interestingly, the prevailing inconsistency in time-course transcriptomics experiments might suggest that transcript rhythmicity is rather arbitrary and is not the central drivers for cell function rhythmicity.**

Another major limitations of this study is the availability of datasets with sufficient replicates, which is more prominent in the plants system. In the mouse system, a few datasets were suitable for our analysis. However, the statistical limitation of the Cry1/2 KO dataset (n=2, Weger et al., 2021) should be acknowledged. As for the plants system, none of the time-course RNA-seq datasets on **truly** arrhythmic mutants found possess enough number of replicates. Therefore, our investigation and conclusions were only based on the mouse model system and new datasets are needed to test this hypothesis in plant systems.

If time allows, further refinement of the pipeline is definitely beneficial. Firstly, ideally all datasets should be processed from FastQ files, with a consistent pipeline for quality control, trimming and (pseudo-)alignment. Secondly, while JTK_CYCLE and LimoRhyde are robust and widely used methods, their assumption of symmetric waveform might make the analysis less accurate. Other rhythmicity detection methods that detect asymmetric waveforms (e.g. empirical JTK_CYCLE) or handle variable sampling duration and frequency (e.g. MetaCycle) could therefore be explored. LimoRhyde2, which has no sinusoidal assumption, is also **worth trying**.

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